Daniel Yao

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Education

Johns Hopkins University Baltimore, MD
M.S.E. Applied Mathematics and Statistics Expected May 2027
4.00 GPA

B.S. Biomedical Engineering, B.S. Applied Mathematics and Statistics Expected May 2027
4.00 GPA, 36 ACT, 1590 SAT, 528 MCAT

Experience

Oberst Lab, Johns Hopkins University Aug 2025 - Present
Undergraduate Research Assistant

- Write and be awarded $3,000, $6,000 Provost's Undergraduate Research Award (2025, 2026)
- Characterize asymptotic properties of active sampling estimators for label-efficient machine learning evaluation
- Investigate heuristic sampling rules for text- and image-based models with Monte Carlo simulations in Python

Johns Hopkins University Aug 2024 - Present
Teaching Assistant

- Lead recitations and hold office hours for upper-level Probability (FA24, SP25, FA25, SP26)
- Hold office hours for Data Structures (FA25, SP26)

Pediatric ICU, Johns Hopkins Medicine Aug 2024 - Present
Undergraduate Research Assistant

- Co-write and be awarded $50,000 Malone Seed Grant (2025) for interdiscplinary research in healthcare
- Compare sedation-agitation scores with vitals, accelerometry, and drug administrations to develop statistical models for pediatric sedation

Clark Lab, Johns Hopkins University Jan 2025 - May 2025
Undergraduate Research Assistant

- Designed reinforcement learning agent with deep Q-learning to regulate pressure-control ventilation in ARDS patients using Gymnasium and PyTorch to select optimal parameters with 97.5% accuracy
- Simulated pressure-volume loop with nonlinear circuit model using PSpice and Simulink to generate data for 17,280 combinations of parameters

Abstracts

Yao, D., Lebarbenchon, K., Clark, A. (2026). Intelligent pressure control ventilation management from a PV loop perspective. Mid-Atlantic Research Exchange. [Best poster award]

Yao, D., Oberst, M. (2026). Label-efficient recall evaluation for zero-shot classification. Mid-Atlantic Research Exchange.

Hoffmann, J., Raghavan, S., Day, M., **Yao, D.**, Morrissey, M, Roy, S., Brown, K., Durr, N., Kudchadkar, S., Fackler, J., LaRosa, J. (2026). Continuous physiological monitoring reveals poor PRN sedation efficacy in pediatric critical care. Critical Care Congress. [Oral presentation]

Raghavan, S., Hoffmann, J., Day, M., Roy, S., Pejic, B., Morrissey, O., Moyer, C., **Yao, D.**, Brown, K., Durr, N., Kudchadkar, S., Fackler, J., LaRosa, J. (2026). Rethinking pediatric sedation assessment: a statistical evaluation of the State Behavioral Scale. Critical Care Congress. [Oral presentation]

Liu, S., Sargent C., Broman L., **Yao, D.** (2024). Role of CRF1 and CRF2 receptors in stress-induced increase in intestinal permeability in the mouse colon. Physiology 39(S1), 815. doi.org/10.1152/physiol.2024.39.S1.815.